

THOMAS TRAN

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EDUCATION

University of Michigan College of Engineering

Ann Arbor, MI

Bachelors of Science in Computer Engineering

Aug. 2023 - Dec. 2025

- **Selected Coursework:** Introduction to Logic Design, Computer Organization, Introduction to Electronic Circuits, Introduction to Signals and Systems, Programming and Intro Data Structures, Digital Integrated Circuits
- **Campus Involvement:** Michigan Aeronautical Science Association(MASA) Avionics Project Member

University of Michigan Dearborn

Dearborn, MI

Bachelors of Science in Computer Engineering

Aug. 2021 - Aug. 2023

TECHNICAL SKILLS

Languages: C++ 20, C 17, Python 3.11, Verilog, ARM, X86

Software: Microsoft Office, Cadence Virtuoso, SPICE, CAD, Matlab, Matlab Simulink, Quartus, ModelSim, Altium

Libraries, Tools, Technologies: CMake, GDB, JSON, openFrameworks, Linux, Git, MacOS, Windows

PROJECTS

Register-transfer level Four Function Calculator - [LINK](#)

- Engineered a Verilog-based four-function calculator on a DE-115 board, processing 11-bit two's complement integers
- Implemented input by using push-buttons and switches for basic arithmetic, with outputs displayed on 7-segment displays, achieving operational accuracy through verification on hardware testing and ModelSim simulations

Traffic Light Controller - [LINK](#)

- Designed a complex traffic light controller system in Verilog, managing multiple scenarios and sensor inputs
- Constructed a finite state machine (FSM) to handle various traffic conditions and timing logic
- Optimized timing logic to control light durations and prevent traffic conflicts, improving overall system efficiency

Vehicle Cruise Control Simulation

- Developed a Simulink-based simulation of a cruise control system, applying principles of control theory to model vehicle dynamics, achieving performance optimization with minimized steady-state errors and controlled oscillations
- Implemented control methods (Proportional, PI, PID) to analyze system responses under different conditions

Tail Light Controller

- Developed a sequential taillight controller using Quartus Lite, managing four distinct signal patterns
- Optimized state tables and equations using Logic Friday, reducing circuit complexity

Data Memory and Cache Simulator

- Built a CPU cache simulator in C++, accurately modeling processor execution of assembly-language programs
- Implemented a write-back, allocate-on-write cache policy with least-recently-used replacement, optimizing cache access

Mars Rover Project

- Led a team of four to design and construct a rover capable of traversing challenging terrain
- Utilized CAD software to create 3D parts, improving design precision and reducing prototyping time
- Programmed rover functions using Arduino, implementing adaptive control algorithms for various surface types

EXPERIENCE

Michigan Aeronautical Science Association(MASA)

Ann Arbor, MI

Avionics Project Member

Sept. 2024 - Present

- Cooperate in a 5-member Avionics sub-team on a radio project development for rocketry application
- Usage of Altium for design and implementation of communication systems for high-altitude rocket

Kura Sushi Novi

Novi, MI

Server

Jul. 2022 - Mar. 2023

- Collaborated with a diverse 4-member rotating team to exceed monthly sales targets by an average of 5%
- Adapted to peak hour rushes, serving up to 20 tables per day while maintaining quality service standards